In [27]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [28]:
```
df=pd.read_csv('E_Commerce.csv')
df                    #here is df stands for data file
```

Out[28]:

	order_id	order_date	state	zone	category	brand_type	customer_
0	ORD0000001	2023-01-31	West Bengal	East	Fashion	Mass	
1	ORD0000002	2023-01-31	Gujarat	West	Sports & Fitness	Mass	
2	ORD0000003	2023-01-31	Delhi NCR	North	Grocery Essentials	Mass	
3	ORD0000004	2023-01-31	Madhya Pradesh	Central	Footwear	Mass	
4	ORD0000005	2023-01-31	Haryana	North	Fashion	Premium	
...	...	...	...	...	...	...	
30595	ORD0030596	2025-12-31	Uttar Pradesh	North	Premium Lifestyle	Mass	
30596	ORD0030597	2025-12-31	Haryana	North	Footwear	Premium	
30597	ORD0030598	2025-12-31	Odisha	East	Footwear	Mass	
30598	ORD0030599	2025-12-31	West Bengal	East	Sports & Fitness	Mass	
30599	ORD0030600	2025-12-31	Maharashtra	West	Footwear	Premium	

30600 rows × 16 columns

In [29]:
```
df.shape
```

Out[29]:
```
(30600, 16)
```

In [30]:
```
nc = list(df.select_dtypes(include=['number']).columns)
nc
```

Out[30]:
```
['customer_age',
 'base_price',
 'discount_percent',
 'final_price',
 'units_sold',
 'revenue']
```

In [31]:
```
df.nunique()
```

Out[31]:
```
order_id                30600
order_date                 36
state                      14
zone                        5
category                    8
brand_type                  2
customer_gender             2
customer_age               48
base_price              29997
discount_percent         5976
final_price             29649
units_sold                146
revenue                 30509
sales_event                 2
competition_intensity       3
inventory_pressure          2
dtype: int64
```

In [32]:
```
df.describe()
```

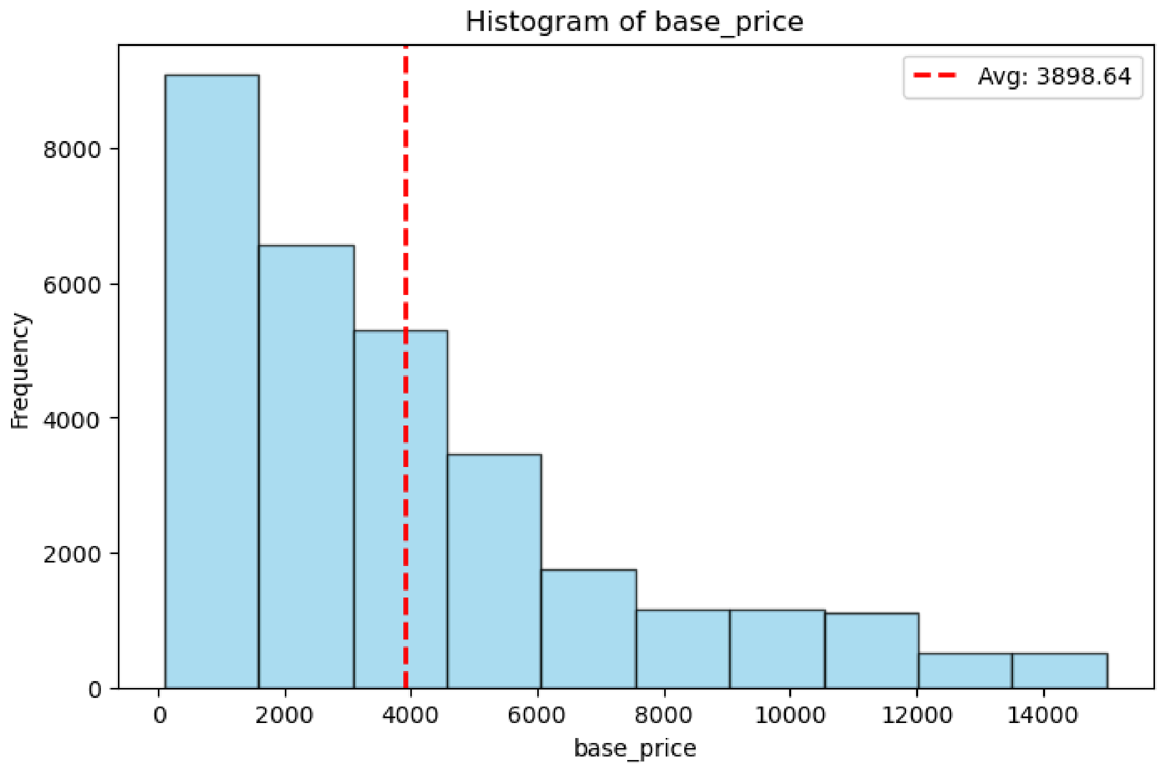
Out[32]:

	customer_age	base_price	discount_percent	final_price	units_sold	
count	30600.000000	30600.000000	30600.000000	30600.000000	30600.000000	306
mean	35.568268	3898.637966	38.111218	2415.015197	31.705490	710
std	12.872780	3371.290126	16.608485	2264.892146	21.381121	836
min	18.000000	100.030000	0.010000	35.010000	2.000000	2
25%	25.000000	1314.637500	24.840000	747.010000	16.000000	166
50%	33.000000	2975.015000	37.290000	1728.605000	27.000000	421
75%	45.000000	5342.480000	51.840000	3309.630000	42.000000	925
max	65.000000	14999.170000	65.000000	14682.280000	161.000000	9061

What is the distribution of average base price in the dataset?

In [33]:
```
ABP = df['base_price'].mean()       #ABP is average base price
print(f"Average base price: {ABP:.2f}")
plt.figure(figsize=(8,5))
plt.hist(df['base_price'], color='skyblue', edgecolor='black', alpha=0.7)
plt.axvline(ABP, color='red', linestyle='dashed', linewidth=2, label=f'Avg: {ABP
plt.xlabel("base_price")
plt.ylabel("Frequency")
plt.title("Histogram of base_price")
plt.legend()
plt.show()
```

```
Average base price: 3898.64
```



In [34]:
```
#separating male and female datasets based on the 'gender' category
Male_data=df[df['customer_gender'] == 'Male']
Female_data=df[df['customer_gender'] == 'Female']
print("Male dataset:\n")
print(Male_data)

print("Female dataset:\n")
print(Female_data)
```

```
Male dataset:

         order_id  order_date          state   zone           category  \
0      ORD0000001  2023-01-31    West Bengal   East            Fashion
1      ORD0000002  2023-01-31        Gujarat   West   Sports & Fitness
2      ORD0000003  2023-01-31      Delhi NCR  North  Grocery Essentials
7      ORD0000008  2023-01-31    West Bengal   East            Fashion
8      ORD0000009  2023-01-31    West Bengal   East  Grocery Essentials
...           ...         ...            ...    ...                ...
30592  ORD0030593  2025-12-31         Odisha   East   Premium Lifestyle
30593  ORD0030594  2025-12-31      Karnataka  South   Premium Lifestyle
30595  ORD0030596  2025-12-31  Uttar Pradesh  North   Premium Lifestyle
30596  ORD0030597  2025-12-31        Haryana  North           Footwear
30599  ORD0030600  2025-12-31    Maharashtra   West           Footwear

      brand_type customer_gender  customer_age  base_price  discount_percent  \
0           Mass            Male            28     1810.89             65.00
1           Mass            Male            19     5678.15             50.83
2           Mass            Male            25      169.98             35.26
7           Mass            Male            23      580.72             46.86
8           Mass            Male            45      132.67             65.00
...          ...             ...           ...         ...               ...
30592    Premium            Male            41     9511.67             39.29
30593       Mass            Male            20     7882.46             53.00
30595       Mass            Male            54    12463.98             38.99
30596    Premium            Male            18     3974.58             39.35
30599    Premium            Male            21      824.75             24.21

       final_price  units_sold    revenue sales_event competition_intensity  \
0           633.81          29   18380.49      Normal                Medium
1          2791.95          67  187060.65      Normal                Medium
2           110.05          57    6272.85      Normal                Medium
7           308.59          27    8331.93      Normal                Medium
8            46.43          52    2414.36    Festival                  High
...            ...         ...        ...         ...                   ...
30592      5774.53          47  271402.91    Festival                Medium
30593      3704.76          45  166714.20      Normal                   Low
30595      7604.27          11   83646.97      Normal                Medium
30596      2410.58          20   48211.60    Festival                   Low
30599       625.08          16   10001.28      Normal                Medium

      inventory_pressure
0                   High
1                   High
2                    Low
7                   High
8                    Low
...                  ...
30592                Low
30593                Low
30595                Low
30596                Low
30599                Low

[15332 rows x 16 columns]
Female dataset:

         order_id  order_date           state     zone  \
3      ORD0000004  2023-01-31  Madhya Pradesh  Central
4      ORD0000005  2023-01-31         Haryana    North
```

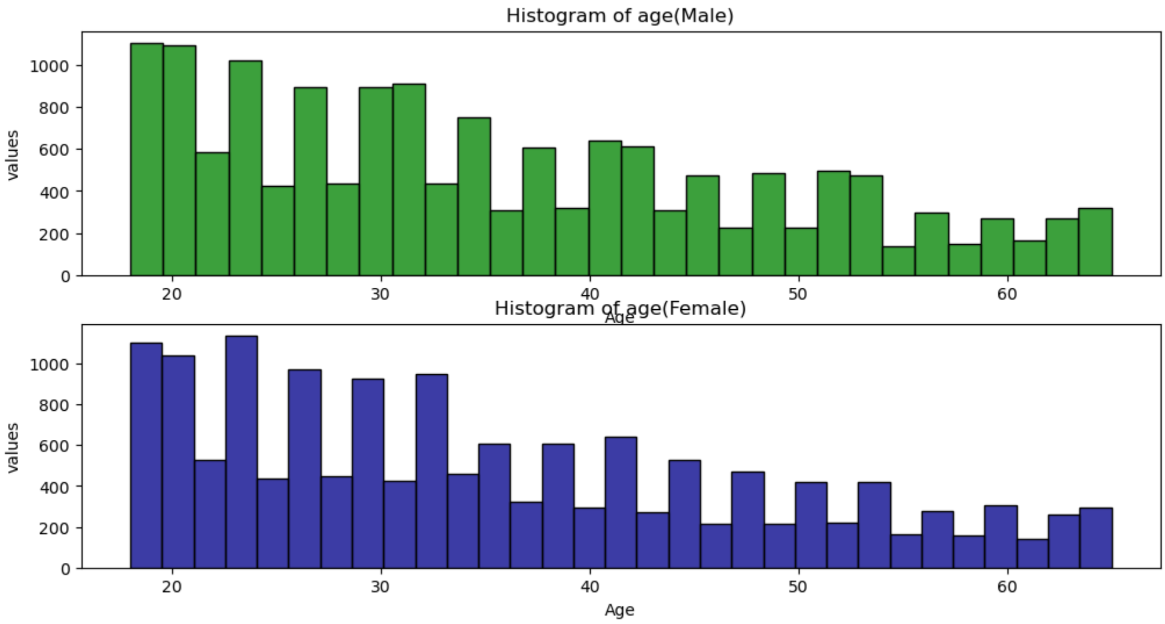
```
5      ORD0000006  2023-01-31          Punjab    North
6      ORD0000007  2023-01-31       Delhi NCR    North
10     ORD0000011  2023-01-31   Uttar Pradesh    North
...           ...         ...             ...      ...
30583  ORD0030584  2025-12-31       Telangana    South
30590  ORD0030591  2025-12-31       Karnataka    South
30594  ORD0030595  2025-12-31  Madhya Pradesh  Central
30597  ORD0030598  2025-12-31          Odisha     East
30598  ORD0030599  2025-12-31     West Bengal     East

                     category brand_type customer_gender  customer_age  \
3                    Footwear       Mass          Female            30
4                     Fashion    Premium          Female            33
5          Grocery Essentials       Mass          Female            19
6               Home & Living       Mass          Female            47
10     Beauty & Personal Care       Mass          Female            22
...                       ...        ...             ...           ...
30583       Premium Lifestyle    Premium          Female            30
30590        Sports & Fitness       Mass          Female            64
30594           Home & Living       Mass          Female            41
30597                Footwear       Mass          Female            58
30598        Sports & Fitness       Mass          Female            23

       base_price  discount_percent  final_price  units_sold    revenue  \
3         3244.91             61.94      1235.01          55   67925.55
4          697.44             48.27       360.79          29   10462.91
5          100.55             65.00        35.19          16     563.04
6         2382.06             46.86      1265.83          34   43038.22
10        1141.42             64.08       410.00          13    5330.00
...           ...               ...          ...         ...        ...
30583     6673.96             42.33      3848.87          21   80826.27
30590     2737.75             65.00       958.21          18   17247.78
30594     3755.81             37.83      2334.99          43  100404.57
30597     3020.42             65.00      1057.15          59   62371.85
30598     1212.05             12.35      1062.36           8    8498.88

      sales_event competition_intensity inventory_pressure
3          Normal                  High               High
4        Festival                Medium               High
5        Festival                  High               High
6          Normal                Medium               High
10         Normal                Medium                Low
...           ...                   ...                ...
30583    Festival                Medium                Low
30590      Normal                  High               High
30594    Festival                Medium               High
30597    Festival                  High                Low
30598      Normal                Medium                Low

[15268 rows x 16 columns]
```

What is the distribution of Male age and Female age data

In [35]:
```
df_male = df[df['customer_gender'] == 'Male']
df_female = df[df['customer_gender'] == 'Female']
plt.figure(figsize=(12,6))
plt.subplot(2,1,1)        #2rows 1 column and 1 is for male data and 2 is female
```

```
sns.histplot(df_male['customer_age'], color='green')
plt.title('Histogram of age(Male)')
plt.xlabel('Age')
plt.ylabel('values')
plt.subplot(2,1,2)
sns.histplot(df_female['customer_age'], color='darkblue')
plt.title('Histogram of age(Female)')
plt.xlabel('Age')
plt.ylabel('values')
```

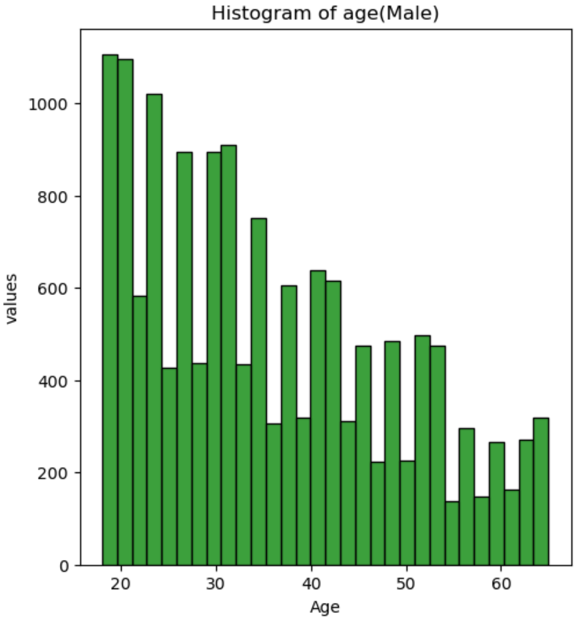
Out[35]: Text(0, 0.5, 'values')

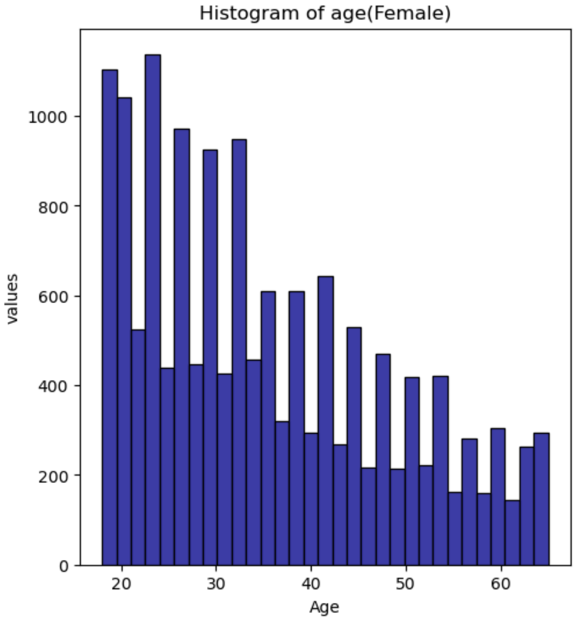


In [36]:

```
#To get 1row 2columns
df_male = df[df['customer_gender'] == 'Male']
df_female = df[df['customer_gender'] == 'Female']
plt.figure(figsize=(12,6))
plt.subplot(1,2,1)
sns.histplot(df_male['customer_age'], color='green')
plt.title('Histogram of age(Male)')
plt.xlabel('Age')
plt.ylabel('values')
plt.subplot(1,2,2)
sns.histplot(df_female['customer_age'], color='darkblue')
plt.title('Histogram of age(Female)')
plt.xlabel('Age')
plt.ylabel('values')
```

Out[36]: Text(0, 0.5, 'values')





Separate Normal and festival sales event

In [37]:
```
#separating sales event datasets based on the 'sales event' category
Normal_data=df[df['sales_event'] == 'Normal']
Festival_data=df[df['sales_event'] == 'Festival']
print("Normal dataset:\n")
print(Normal_data)

print("Festival dataset:\n")
print(Festival_data)
```

```
Normal dataset:

        order_id  order_date           state     zone            category  \
0      ORD0000001  2023-01-31     West Bengal     East             Fashion
1      ORD0000002  2023-01-31         Gujarat     West    Sports & Fitness
2      ORD0000003  2023-01-31       Delhi NCR    North  Grocery Essentials
3      ORD0000004  2023-01-31  Madhya Pradesh  Central            Footwear
6      ORD0000007  2023-01-31       Delhi NCR    North       Home & Living
...           ...         ...             ...      ...                 ...
30591  ORD0030592  2025-12-31     Maharashtra     West  Grocery Essentials
30593  ORD0030594  2025-12-31       Karnataka    South   Premium Lifestyle
30595  ORD0030596  2025-12-31   Uttar Pradesh    North   Premium Lifestyle
30598  ORD0030599  2025-12-31     West Bengal     East    Sports & Fitness
30599  ORD0030600  2025-12-31     Maharashtra     West            Footwear

      brand_type customer_gender  customer_age  base_price  discount_percent  \
0           Mass            Male            28     1810.89             65.00
1           Mass            Male            19     5678.15             50.83
2           Mass            Male            25      169.98             35.26
3           Mass          Female            30     3244.91             61.94
6           Mass          Female            47     2382.06             46.86
...          ...             ...           ...         ...               ...
30591       Mass            Male            26      489.74             20.55
30593       Mass            Male            20     7882.46             53.00
30595       Mass            Male            54    12463.98             38.99
30598       Mass          Female            23     1212.05             12.35
30599    Premium            Male            21      824.75             24.21

       final_price  units_sold    revenue sales_event competition_intensity  \
0           633.81          29   18380.49      Normal                Medium
1          2791.95          67  187060.65      Normal                Medium
2           110.05          57    6272.85      Normal                Medium
3          1235.01          55   67925.55      Normal                  High
6          1265.83          34   43038.22      Normal                Medium
...            ...         ...        ...         ...                   ...
30591       389.10          28   10894.80      Normal                  High
30593      3704.76          45  166714.20      Normal                   Low
30595      7604.27          11   83646.97      Normal                Medium
30598      1062.36           8    8498.88      Normal                Medium
30599       625.08          16   10001.28      Normal                Medium

      inventory_pressure
0                   High
1                   High
2                    Low
3                   High
6                   High
...                  ...
30591                Low
30593                Low
30595                Low
30598                Low
30599                Low

[22889 rows x 16 columns]
Festival dataset:

        order_id  order_date           state     zone            category  \
4      ORD0000005  2023-01-31         Haryana    North             Fashion
5      ORD0000006  2023-01-31          Punjab    North  Grocery Essentials
```

```
8      ORD0000009  2023-01-31     West Bengal     East  Grocery Essentials
11     ORD0000012  2023-01-31          Kerala    South  Grocery Essentials
14     ORD0000015  2023-01-31          Odisha     East       Home & Living
...           ...         ...             ...      ...                 ...
30583  ORD0030584  2025-12-31       Telangana    South   Premium Lifestyle
30592  ORD0030593  2025-12-31          Odisha     East   Premium Lifestyle
30594  ORD0030595  2025-12-31  Madhya Pradesh  Central       Home & Living
30596  ORD0030597  2025-12-31         Haryana    North            Footwear
30597  ORD0030598  2025-12-31          Odisha     East            Footwear

      brand_type customer_gender  customer_age  base_price  discount_percent  \
4        Premium          Female            33      697.44             48.27
5           Mass          Female            19      100.55             65.00
8           Mass            Male            45      132.67             65.00
11       Premium            Male            21      701.96             43.81
14       Premium            Male            53     2000.65             18.44
...          ...             ...           ...         ...               ...
30583    Premium          Female            30     6673.96             42.33
30592    Premium            Male            41     9511.67             39.29
30594       Mass          Female            41     3755.81             37.83
30596    Premium            Male            18     3974.58             39.35
30597       Mass          Female            58     3020.42             65.00

       final_price  units_sold    revenue sales_event competition_intensity  \
4           360.79          29   10462.91    Festival                Medium
5            35.19          16     563.04    Festival                  High
8            46.43          52    2414.36    Festival                  High
11          394.43          49   19327.07    Festival                  High
14         1631.73          14   22844.22    Festival                   Low
...            ...         ...        ...         ...                   ...
30583      3848.87          21   80826.27    Festival                Medium
30592      5774.53          47  271402.91    Festival                Medium
30594      2334.99          43  100404.57    Festival                Medium
30596      2410.58          20   48211.60    Festival                   Low
30597      1057.15          59   62371.85    Festival                  High

      inventory_pressure
4                   High
5                   High
8                    Low
11                   Low
14                   Low
...                  ...
30583                Low
30592                Low
30594               High
30596                Low
30597                Low

[7711 rows x 16 columns]
```

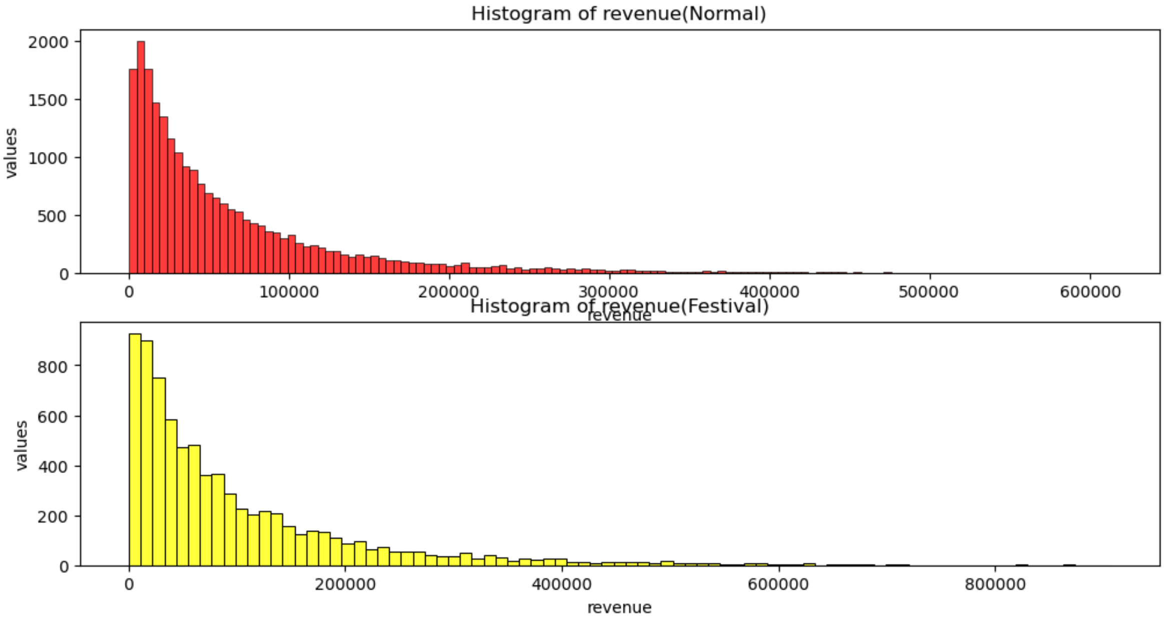
What is the distribution of Normal sales revenue and festival sales revenue

In [38]:

```
df_normal = df[df['sales_event'] == 'Normal']
df_festival = df[df['sales_event'] == 'Festival']
plt.figure(figsize=(12,6))
plt.subplot(2,1,1)       #2rows 1 column and 1 is for male data and 2 is female
```

```
sns.histplot(df_normal['revenue'], color='red')
plt.title('Histogram of revenue(Normal)')
plt.xlabel('revenue')
plt.ylabel('values')
plt.subplot(2,1,2)
sns.histplot(df_festival['revenue'], color='yellow')
plt.title('Histogram of revenue(Festival)')
plt.xlabel('revenue')
plt.ylabel('values')
```

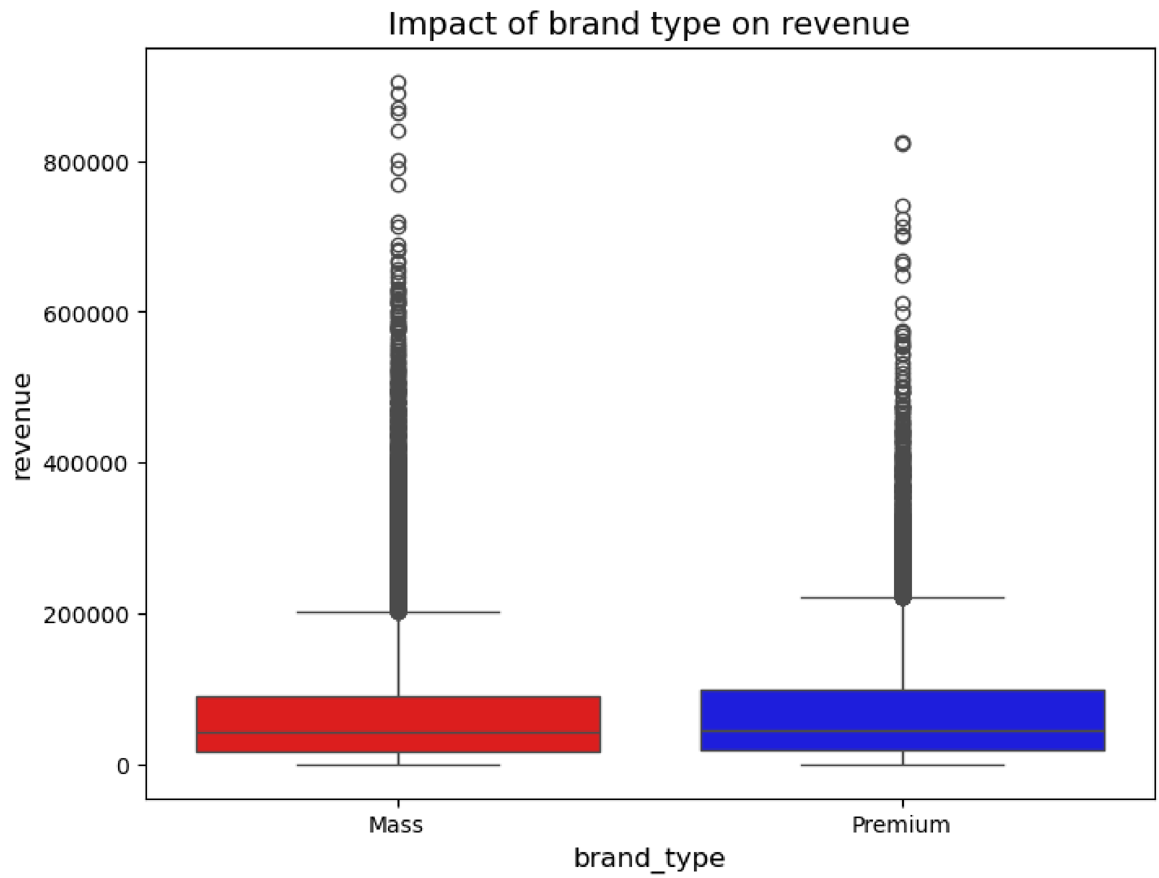
Out[38]: `Text(0, 0.5, 'values')`



How does brand type affects revenue(box plot)

In [39]:
```
plt.figure(figsize=(8,6))
sns.boxplot(x='brand_type', y='revenue', data=df, palette=['red', 'blue'])
plt.xlabel("brand_type", fontsize=12)
plt.ylabel("revenue", fontsize=12)
plt.title("Impact of brand type on revenue", fontsize=14)
plt.show()
```

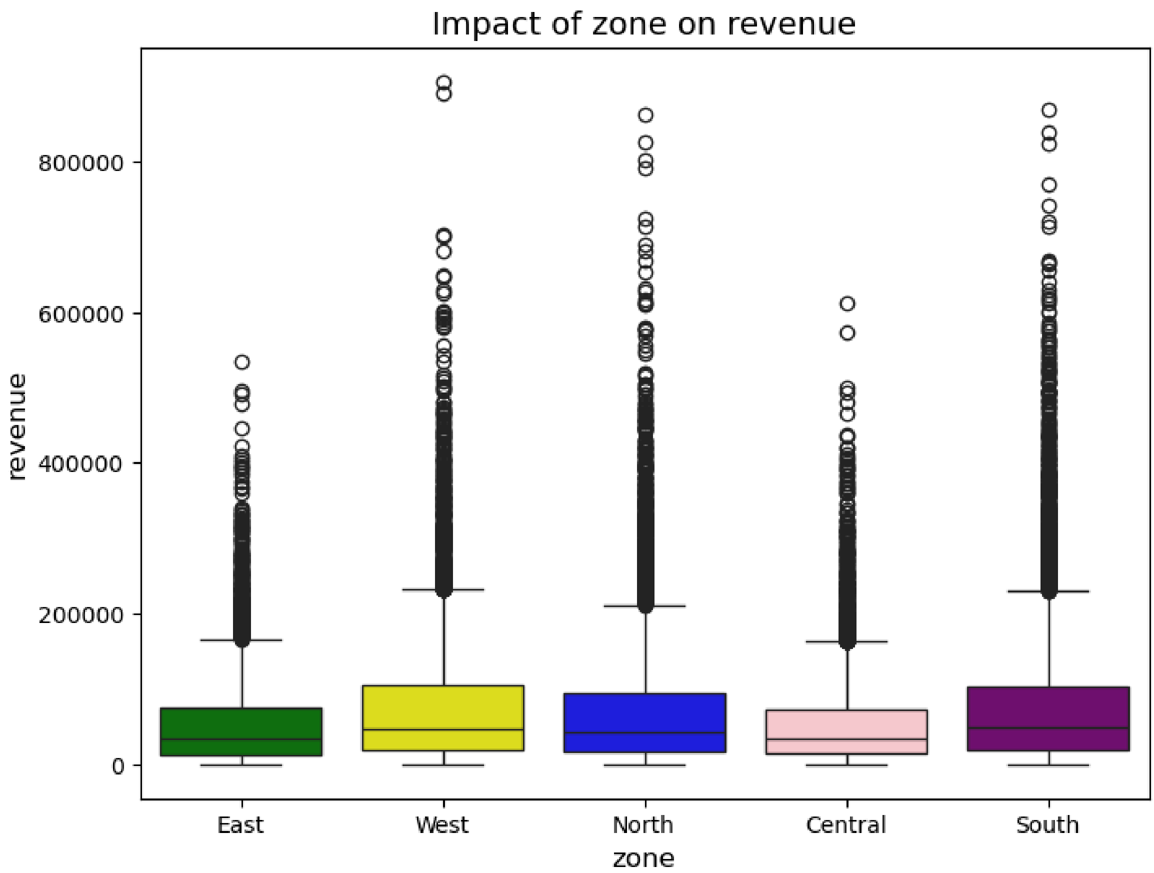
```
C:\Users\ADMIN\AppData\Local\Temp\ipykernel_10588\1117936685.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.boxplot(x='brand_type', y='revenue', data=df, palette=['red', 'blue'])
```



How does zone affects revenue (Box plot)

In [41]:
```
plt.figure(figsize=(8,6))
sns.boxplot(x='zone', y='revenue', data=df, palette=['green', 'yellow', 'blue',
plt.xlabel("zone", fontsize=12)
plt.ylabel("revenue", fontsize=12)
plt.title("Impact of zone on revenue", fontsize=14)
plt.show()
```

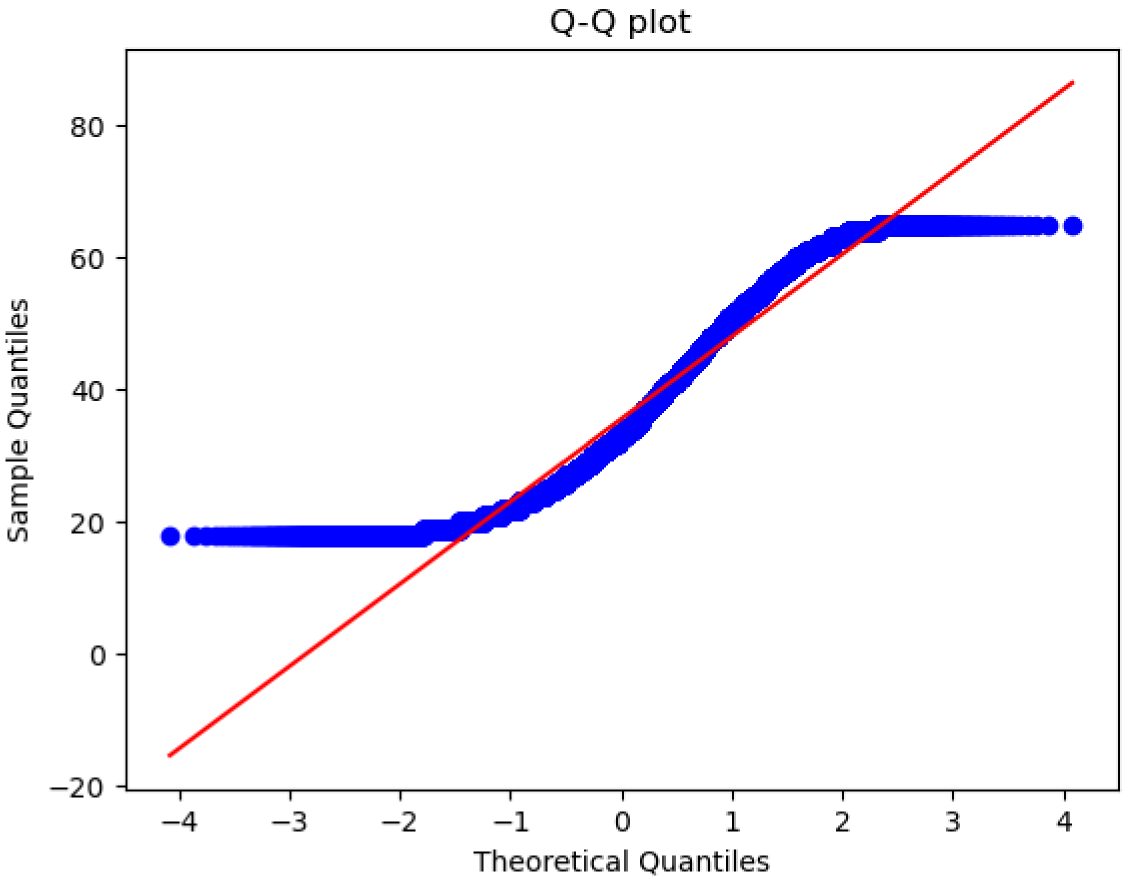
```
C:\Users\ADMIN\AppData\Local\Temp\ipykernel_10588\263941267.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.boxplot(x='zone', y='revenue', data=df, palette=['green', 'yellow', 'blue',
'pink', 'purple'])
```



Comment on normality of customer age by constructing a QQ plot

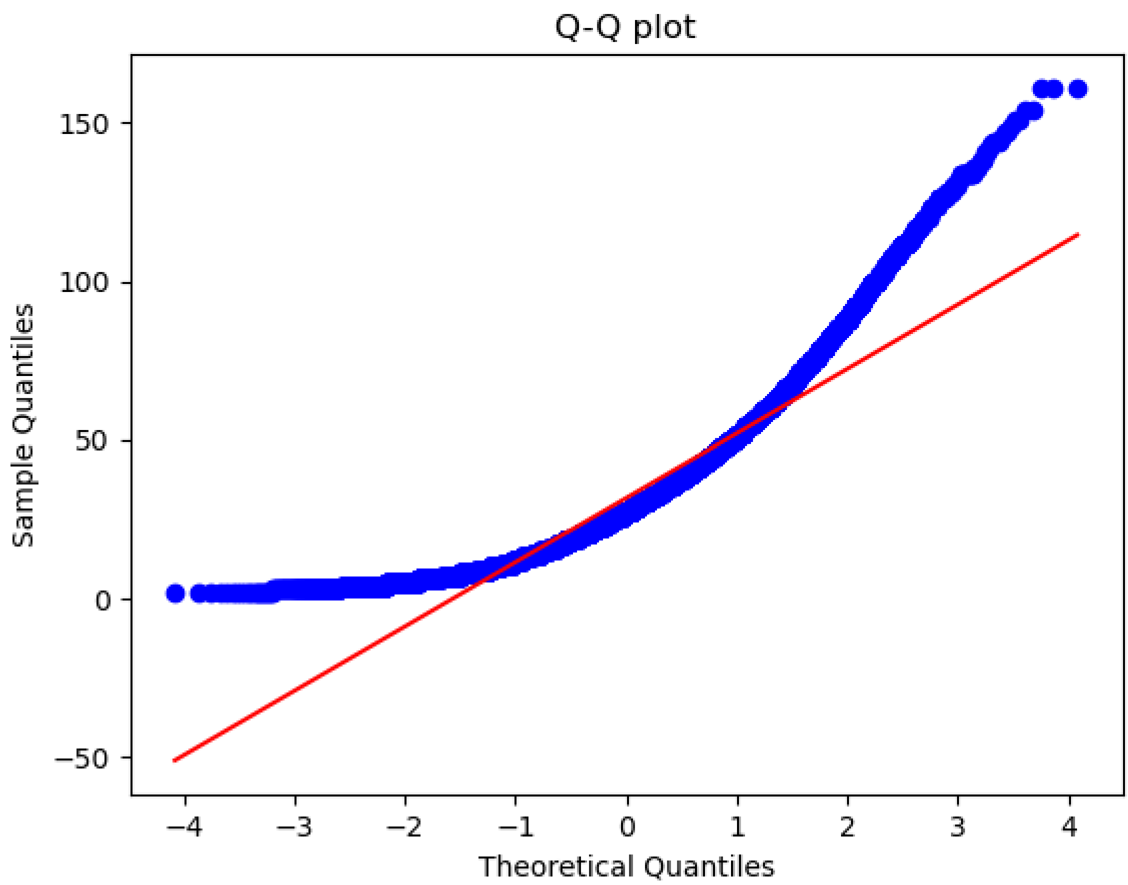
In [42]:
```
from scipy import stats
```

In [44]:
```
df1 = df['customer_age']
stats.probplot(df1, dist="norm", plot=plt)
plt.xlabel('Theoretical Quantiles')
plt.ylabel('Sample Quantiles')
plt.title('Q-Q plot')
plt.show()
```



Comment on normality of units sold by constructing a QQ plot

In [45]:
```
df2 = df['units_sold']
stats.probplot(df2, dist="norm", plot=plt)
plt.xlabel('Theoretical Quantiles')
plt.ylabel('Sample Quantiles')
plt.title('Q-Q plot')
plt.show()
```



positively skewed

In [ ]:

```

```